HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented mechanical components for European air-cooled chiller systems, integrating fluid, thermal, and refrigeration principles into detailed design work.
- Developed and planned project activities for assigned mechanical components, translating conceptual design ideas into product specifications and release documentation.
- Selected and specified mechanical components including heat exchangers, fans, compressors, and valves, balancing performance, cost, and reliability requirements for new product development.
- Analyzed and troubleshot mechanical sub-components based on test data from validation programs, modifying designs to enhance performance and reliability.
- Collaborated with manufacturing and design teams to define layout concepts and system configurations, ensuring alignment with cost, performance, and reliability targets.
- Applied European mechanical and safety codes, including CE marking, the Pressure Equipment Directive (PED), and harmonized standards like EN 378 and EN 14511, to product designs.
- Containerized Java Spring Boot microservices for a payments operations dashboard using **Docker** and deployed them to an **AWS** EKS cluster, improving service resilience.

SKILLS

Engineering & Design – Mechanical Design, Fluid Dynamics, Thermal Principles, Refrigeration Systems, Component Selection, Design Reviews, CE Marking, PED, Machinery Directive, EN 378, EN 14511

Software Development – Java, Spring Boot, REST APIs, Microservices, Python, SQL, PostgreSQL, Docker, AWS (EC2, S3, Lambda, CloudWatch), Git, Jenkins

Data Analysis & Reporting – Data Analysis, Test Data Analysis, Product Specifications, Documentation Release, Power BI, Tableau, R, pandas, NumPy, scikit-learn, DAX

Project Management & Collaboration – Project Planning, Task Definition, Team Collaboration, Cross-functional Support, Mentoring, Technical Documentation, Change Management, Agile Methodologies

Operating Systems & Tools – Linux, Shell Scripting, Cron Jobs, Jira, Confluence, Splunk, Dynatrace

PROJECTS

Chiller System Performance Optimization (HCL Technologies)

- Engineered Python scripts to automate the parsing and analysis of large datasets from component testing, identifying performance bottlenecks in refrigeration cycles.
- Developed automated data validation routines using **SQL** and **Python** to ensure the integrity of test results for over **1,500** individual component tests.
- Created interactive **Power BI** dashboards to visualize performance metrics and failure modes, enabling engineering teams to quickly identify areas for design improvement.

Enterprise Java Application Modernization (HCL Technologies)

- Modernized legacy Java components to **Spring Boot** microservices, enhancing system modularity and maintainability across the application.
- Implemented robust **REST APIs** using **Spring Boot** for inter-service communication, handling approximately **500** requests per second during peak loads.
- Integrated **Hibernate** with PostgreSQL for data persistence, managing complex relational data models and ensuring transactional integrity for core business logic.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)